Figure 2 shows the mean atmospheric temperature within a 20-mile radius of the Mount Pinatubo volcano in the Philippines between 1990 and 2000. Mount Pinatubo erupted in 1991.

Figure 2

Year	Mean atmospheric temperature (°C)
1990	15.1
1991	14.7
1992	14.8
1993	14.7
1994	14.7
1995	14.6
1996	14.7
1997	14.8
1998	14.7
1999	14.8
2000	14.9

(a) Tick (✓) the box that **best** describes the change in carbon dioxide concentration in the atmosphere between 1930 and 2010. [1]

Carbon dioxide concentration increased by approximately 10 ppm every 10 years ☐

Carbon dioxide concentration increased more between 1970 and 2010 than it did between 1930 and 1960 ☐

Carbon dioxide concentration increased more between 1930 and 1960 than it did between 1970 and 2010 ☐

There is no trend to the change in carbon dioxide concentration between 1930 and 2010 ☐

Examiner only

(b) A newspaper report states that global warming is caused by increases in atmospheric carbon dioxide levels **and** by increases in solar activity.

Explain whether the data in **Figure 1** supports this claim. [2]

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(c) **Figure 2** shows that the mean atmospheric temperature near Mount Pinatubo decreased after the volcanic eruption of 1991. Suggest why scientists would **not** use these findings to explain how volcanic activity affects **global** temperatures. [1]

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(d) State **one** industrial method that is being developed to reduce the concentration of carbon dioxide in the atmosphere. [1]

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(e) The Earth's original atmosphere contained about 95% carbon dioxide. Today, the atmosphere contains 0.04% carbon dioxide. Explain why the level of carbon dioxide decreased over geological time. [3]

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(f) When carbon dioxide turns limewater milky, calcium hydroxide is produced.

Give the formula of calcium hydroxide. [1]

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